

Designing with the Mind in Mind

The Psychological Basis of User Interface Design Guidelines

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ABSTRACT

UI design rules and guidelines are not simple recipes. Applying them effectively requires determining rule applicability and precedence and balancing trade-offs when rules compete. By understanding the underlying psychology, designers and evaluators enhance their ability to apply design rules. This two-part (160-minute) course explains that psychology.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design theory, concepts and paradigms; Interaction design; Interaction design process and methods; User interface design; Interaction design; Interaction design process and methods; User centered design; Visualization; Visualization theory, concepts and paradigms; Accessibility; Accessibility theory, concepts and paradigms.

KEYWORDS

perception, cognition, psychology, design heuristics, theory

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1 BENEFITS

User interface design principles, guidelines, and heuristics (collectively called “design guidelines”) have a basis in the psychology of human perception, learning, memory, and problem-solving. Unfortunately, people who design and evaluate user interfaces usually learn the design rules without understanding their psychological basis.

UI design guidelines are not simple recipes to be applied mindlessly. Applying them effectively requires determining their applicability (and precedence) in specific situations. It also requires balancing the trade-offs that inevitably arise in situations when design rules appear to contradict each other.

By understanding the underlying psychology for the design rules, designers and evaluators enhance their ability to interpret and apply

them. Explaining that psychology is the focus of this course. The first part focuses on perception; the second focuses on cognition.

1.1 Learning Objectives

After taking this course, attendees will understand the basic perceptual and cognitive psychology underlying UI design principles, guidelines, and heuristics. The hoped-for outcome is that the usability of interactive systems will be improved.

2 INTENDED AUDIENCE

This course is intended for software designers and developers of all experience levels, especially those who did not take perceptual and cognitive psychology in college or who want to update their knowledge of it. Others who may benefit: Software Q/A engineers, usability testers, development managers.

3 PREREQUISITES

None.

4 CONTENT

This course covers the following topics (with approximate times):

Introduction (5 mins)

Perception (75 mins)

- Perception is biased by experience, context, goals
- Vision is optimized to perceive structure (Gestalt principles)
- We seek and use structure
- Color vision is limited
- Peripheral vision is poor, and visual search is linear unless target “pops” in periphery

Cognition (75 mins)

- Reading is unnatural
- Attention is limited; Memory is imperfect
- Limits on attention and memory shape our thought and action, e.g., change-blindness
- Recognition is easier than recall
- System 1 vs. System 2 (per Kahneman, *Thinking Fast & Slow*, 2013)
- Easy: learning from experience, and executing learned actions; Hard: novel actions, problem-solving, and calculation
- Hand-eye coordination follows rules
- Human error

Summary, Q&A, wrap-up, evaluations (5 mins)

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5 PRACTICAL WORK

The course includes many engaging visual examples and demonstrations, and brief audience exercises.

6 INSTRUCTOR BACKGROUND

Jeff Johnson is a Professor in the Computer Science Department of the University of San Francisco. After earning B.A. and Ph.D. degrees from Yale and Stanford, he worked at Cromemco, Xerox, US West, Hewlett-Packard Labs, and Sun Microsystems. In 1990, he co-chaired the first Participatory Design conference, PDC'90. He serves on the SIGCHI U.S. Public Policy Committee. He has also taught at Stanford University and Mills College, and in 2006 and 2013 he taught HCI as an Erskine Fellow at the University of

Canterbury in New Zealand. He is a member of the ACM SIGCHI Academy, a SIGCHI Lifetime Achievement in Practice Awardee, and an ACM Distinguished Member. He has authored or co-authored many articles and chapters on Human-Computer Interaction, as well as the books *GUI Bloopers*, *Web Bloopers*, *GUI Bloopers 2.0*, *Designing with the Mind in Mind*, *Conceptual Models* (co-authored with Austin Henderson), *Designing with the Mind in Mind*, 2nd edition, *Designing UIs for an Aging Population* (co-authored with Kate Finn), and *Designing with the Mind in Mind*, 3rd edition.

REFERENCES

- [1] Johnson, J. (2020) *Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines*, 3rd edition, Morgan Kaufmann Publisher.